Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 4×3

2. 4×7

3. 4×4

4. 4×9

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

6. $2.5 + 0.25$

7. 5×5

8. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

9. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

10. Is 6 : 10 equivalent to 9 : 15?

11. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

12. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

13. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

14. Red: blue is 2 : 3. Complete (2, 3), (4,__).

15. A mix has 9 raisins and 12 nuts. Divide both counts by 3; write the new ratio.

16. A recipe uses 8 cups oats for 10 cups seeds. How many cups seeds go with 12 cups oats?

17. Gold: silver beads are 5 : 6. There are 66 beads. Find each count.

18. Orange: purple tiles are 6 : 7. What fraction of all tiles is orange?

CHALLENGE / Explain your plan and check the result

C1. A banner repeats 4 red flags and 7 green flags. You have 28 red and 52 green flags. Make as many complete patterns as possible. How many flags of each color are unused?

C2. Paint uses 4 cups yellow for 6 cups blue. A small batch uses 3 cups yellow. Find the blue amount and total paint. Explain a non-whole scale factor.